

Closed-Loop Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Energy Absorbers

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Tensegrity-inspired architectures, in which rigid struts are suspended within a continuous flexible network, can exhibit tunable nonlinear responses and high strength-to-weight ratios, motivating their use in lightweight energy-absorbing structures. Multi-material fused deposition modeling (FDM) can co-print rigid PLA struts and flexible TPU elements in a single build, but the resulting design space—spanning strut geometry, tension-element cross-section, topology, and unit-cell tiling—is too large to explore by trial and error.

This work presents a closed-loop experimental campaign using Bayesian optimization (BO) to develop multi-material 3D-printed tensegrity-inspired structures for improved energy absorption under quasi-static compression and drop-weight impact. A Gaussian-process surrogate trained on peak transmitted force, specific energy absorption (SEA), and compaction efficiency recommends each next batch of designs, enabling sequential design selection without finite-element simulation. The workflow combines automated surrogate updating and experiment selection with manual specimen handling and test setup.

We parameterize a family of unit cells in which rigid PLA struts are enclosed by continuous TPU skins to improve interfacial interlocking. Design variables include strut diameter, length, tension-element width and thickness, strut count per cell, and tiling pattern. Specimens are fabricated on a multi-material FDM system and tested under quasi-static compression and instrumented drop-weight impact.

We will report how BO-selected designs evolve across iterations, how surrogate predictions change as data accumulate, and what Pareto-efficient trade-offs emerge between SEA and peak force. We will also discuss practical lessons for the closed-loop workflow, including print-failure handling, batch variability, and exploration–exploitation balance.